



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

AI chip that learns on its own at the edge

BrainChip Holdings has introduced the Akida AKD1500 reference chip, built on a 22nm fully depleted silicon-on-insulator (22FDX) process. Designed for edge AI inference, the chip consumes less than one watt of power and can operate either as a standalone AI processor or as

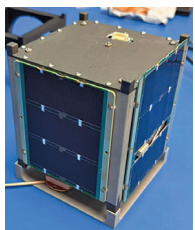


a hardware co-processor enabling AI processing without continuous cloud connectivity. It uses event-driven neuromorphic computing to process data only when events occur rather than continuously, reducing power consumption compared to conventional AI accelerators. It also includes serial interfaces for battery-powered embedded systems, allowing low-cost microcontrollers to add pattern recognition and on-device learning without requiring a system redesign.

BrainChip Holdings
<https://brainchip.com>

World's first commercial nuclear-powered satellite

City Labs has launched the BOHR (betavoltaic orbital high-reliability) satellite, the world's first commercial nuclear-powered satellite and the first CubeSat



powered by nuclear technology. The satellite uses the company's Nano-

Tritium betavoltaic technology, which converts energy from the natural radioactive decay of tritium directly into

electricity using a semiconductor. The power system is designed to operate for many years with little degradation, making it suitable for long-duration missions and operation in dark environments such as permanently shadowed lunar craters, where solar panels cannot generate power.

City Labs
<https://citylabs.net>

Smart driving chip with 1280 TOPS AI computing

Li Auto has unveiled its in-house developed Mach M100 smart driving chip. Built on a 5nm automotive-grade process, the chip delivers up to 1280 TOPS of AI computing performance per chip, which the



company claims is the highest for an automotive AI chip. The chip uses a dataflow-based AI architecture instead of a traditional cache-based design to improve data movement efficiency for AI workloads. The company also developed its VLA 2.1 AI model alongside the M100 chip, reducing end-to-end processing latency by 40% and enabling response speeds it claims are twice as fast as human reaction.

Li Auto
<https://ir.lixiang.com>

Networking without depending on internet or cloud

EVE is a decentralised, local-first networking and computing platform that enables devices to communicate, share data, and run applications without depending on the internet or centralised cloud infrastructure. It creates

peer-to-peer mesh networks in which every device can act as both a client and a node, allowing services to continue operating during internet outages. It supports offline operation with automatic



synchronisation when connectivity becomes available, end-to-end encryption for communication, distributed identity, and authentication without a central authority, and networking that removes single points of failure.

EVE
<https://eve.civoraa.org>

Power management IC with wheel-speed sensing

Allegro MicroSystems has introduced the A81415 power management IC (PMIC). The device combines power management and wheel-speed sensing in a single chip, replacing separate safety PMICs,



wheel-speed decoder ICs, and multiple discrete power components used in conventional designs. According to the company, this reduces the external component count by up to nine, lowers semiconductor bill-of-materials cost by up to \$4 per vehicle, and frees more than 50% of PCB space. The PMIC processes wheel-speed sensor data internally and transfers decoded data to the microcontroller through a serial peripheral interface, reducing processing latency and lowering the microcontroller workload.

Allegro MicroSystems
<https://www.allegromicro.com>